Report Progetto 2 -English

Data Fusion: Camera RGB + LiDAR 3D

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Scope: People detection and depth estimation on the KITTI Raw dataset.
Technology Stack: Python, NumPy, OpenCV, Streamlit, Open3D.

Objective and Motivation

The objective of the project is to combine the images of an RGB camera with the 3D measurements of a LiDAR to identify people within the scene and accurately estimate their depth (distance).
Why resort to Data Fusion? Individual sensors present intrinsic limitations that are compensated by integration:

- **RGB Camera**: Provides colors, textures, and shapes ideal for visually recognizing a person, but does not allow deriving real distance directly from a single photo.
- **3D LiDAR**: Provides precise 3D spatial positions and distances, but interpreting the shape using only the point cloud can be complex (often with few points on the object).
- **RGB + LiDAR**: Associates visual recognition with a precise metric depth position.

Dataset Used

The project is based on a subset of **32 frames from the KITTI Raw** dataset, captured by a moving vehicle. Each frame includes:

- An RGB photo of the scene.
- A LiDAR scan (3D point cloud).
- A common index for data synchronization.

Methodology and Data Processing

- **Geometry and Calibration**: LiDAR points (system with X forward, Y left, Z up) are converted into the camera reference system via the transformation matrix $T_velo_to_cam$. Subsequently, the combined projection matrix M (which includes rectification and intrinsic parameters) projects the 3D points (X, Y, Z) onto the pixel (u, v) coordinates of the image through perspective division $u = a/d$ e $v = b/d$.
- **People Detection (HOG + SVM)**:
  - **HOG (Histogram of Oriented Gradients)**: Extracts descriptive features based on brightness gradients to outline the contours of the human figure.
  - **Linear SVM (Support Vector Machine)**: Classifies image areas to identify the presence of people.
  - **NMS (Non-Maximum Suppression)**: Reduces duplicate boxes on the same subject.
- **Depth Estimation and IQR Filter**: Since points belonging to the background or the road may be present inside the bounding box, the following is applied:
  1. **IQR Filter (Interquartile Range)**: Discards depth outliers, keeping only those included in the interval $[Q1 - 1.5 \cdot IQR,\ Q3 + 1.5 \cdot IQR]$.
  2. **Median**: The median of the filtered depth values is calculated to determine the final distance of the subject.

Practical Application (Streamlit Demo)

The interactive application allows selecting an RGB capture, viewing the LiDAR data, and running detection with the dedicated button to display on screen the recognition boxes, the estimated depth, and a summary table of points and HOG scores.

Future Developments

To increase system performance, three directions for improvement are proposed:

1. **Advanced Detectors**: Replace the HOG + SVM combination with a pre-trained model like YOLO (filtered on the person class) to improve detection on small or partially hidden subjects and reduce false positives.
2. **Segmentation Masks**: Use a precise mask of the person's silhouette instead of the entire rectangular bounding box, reducing the inclusion of road/background points in the depth calculation.
3. **Temporal Tracking**: Introduce tracking to associate the same person across successive frames and filter depth fluctuations over time.